STEFANO CASIRATI, Ph.D.

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RESEARCH INTERESTS

Eco-Hydrological Modeling, Computational Hydrology, Remote Sensing, Forests, Agriculture, Irrigation, Earth System Models, Food Energy Water Systems, Land-Atmosphere Interactions, Climate Teleconnections.

EDUCATION

2023	**Ph.D. in Environmental Systems**	University of California, Merced, California, USA
		School of Engineering,
		Advisors: Dr. Martha Conklin, Dr. Safeeq Khan
2014	**M.Eng. Environmental Engineering,**	Politecnico of Milan, Italy
		Advisors: Dr. Maria Cristina Rulli, Dr. Paolo d'Odorico
2011	**B.Eng. Environmental Engineering,**	Politecnico of Milan, Italy
		Advisor: Dr. Maria Cristina Rulli

PROFESSIONAL EXPERENCE

12/04/2023 - Present	Postdoctoral Researcher, University of California, Berkeley, USA
2015–2018	Suntrading S.p.A. / Sunsaving S.r.l. / Solarfast S.p.A., Italy Consultant, environmental engineer, energy efficiency analyst
2015-2015	Tecnoambiente S.r.l., Italy - Consultant, environmental engineer (HSE)
2014–2015	Department of Civil and Environmental Engineering, Politecnico of Milan, Italy Research internship: *"Virtual water analysis for food and energy";* Project: *"The role of Europe in land deals: understanding the food-energy-water security nexus"* (PI Dr. Maria Cristina Rulli)

PUBLICATIONS

Peer-Reviewed Journal Articles

2026 **Casirati, S.,** D'Odorico P., Fung I., Molod A., and Girotto M. (2026). "Differential impacts of irrigation on precipitation across the United States". *Environmental Research Letters*. https://doi.org/10.1088/1748-9326/ae929d

2026 **Casirati, S.**, M. H. Conklin, and M. Safeeq. (2026). "Hydrological Response to Compounding Impacts of Climate Change and Forest Management in the Upper Kings River Basin, CA, USA." *Ecohydrology* 19, no. 1: e70157. https://doi.org/10.1002/eco.70157

2025 **Casirati, S.**, D'Odorico, P., Reichle, R. H., Rouf, T., Stookey, M., & Girotto, M. (2025). A model-based evaluation of the effects of irrigation expansion on regional and global land surface climate. Earth's Future, 13, e2025EF006271. https://doi.org/10.1029/2025EF006271

Casirati, S., Conklin, M., Nandi, S. and Safeeq, M. (2025), Effect of Forest Management Practices on Water Balance Across a Water–Energy Gradient in the Upper Kings River Basin, USA. Ecohydrology, 18: e2753. https://doi.org/10.1002/eco.2753

2023 **Casirati S**., Conklin M.H. and Safeeq M. (2023) Influence of snowpack on forest water stress in the Sierra Nevada. *Front. For. Glob. Change* 6:1181819. https://doi.org/10.3389/ffgc.2023.1181819

2019 Rulli M.C., **Casirati S.**, Dell'Angelo J., Davis K.F., Passera C., D'Odorico P. (2019) Interdependencies and telecoupling of oil palm expansion at the expense of Indonesian rainforest. *Renewable and Sustainable Energy Reviews* 105, 499-512. https://doi.org/10.1016/j.rser.2018.12.050

Conference Proceedings

2015 D'Odorico P., Rulli M.C., **Casirati S.**, Davis K.F., Dell'Angelo J., Land use change induced by large scale land acquisitions. World Bank Conference on Land and Poverty, Washington DC, 23-27 March 2015.

PRESENTATIONS

2026 **Casirati S.,** D'Odorico P., Fung I., Girotto M., (2026) "Irrigation and Land–Atmosphere Interactions: Insights into Global Climate Patterns", Oral Presentation, 106th AMS American Meteorological Society Annual Meeting, Houston, USA

2025 **Casirati S.,** D'Odorico P., Fung I., Girotto M., (2025) "Irrigation's Global and Regional Climate Impacts: Exploring Land–Atmosphere Feedbacks Across S2S Timescales", Poster Presentation, Advancing Understanding of Land-Atmosphere Interactions and Processes on S2S Predictability Workshop, Boulder, Colorado, USA

Safeeq, M. and **Casirati, S.**, (2025) "Forest Management in a Warming World: Enhancing Insights into Compounding Hydrologic Impacts", Oral Presentation, European Geophysical Union General Assembly 2025, Vienna, Austria.

2024	**Casirati S.,** D'Odorico P., Girotto M., (2024) "Evaluating the effects of irrigation expansion on regional and global land surface climate using a Land Surface Model", Poster Presentation, American Geophysical Union, Fall Meeting. San Francisco, California, USA
	Safeeq, M., **Casirati, S.,** Nandi, S. and Conklin, M., (2024) "Modeling the effects of forest thinning on the long-term water balance in a Mediterranean mixed-conifer basin", Oral Presentation, American Geophysical Union, Fall Meeting. San Francisco, California, USA
2022	**Casirati S.,** Safeeq M., and Conklin M. H., (2022) "Predicting drought-related tree mortality in the Sierra Nevada.", eLightning presentation Frontiers in Hydrology Meeting – San Juan, Puerto Rico, USA
2020	**Casirati S.**, Safeeq M., and Conklin M. H., (2020) "Influence of snow accumulation and ephemerality on recent tree mortality in the Sierra Nevada", Oral Presentation, Yosemite Hydroclimate Meeting, Virtual.
2020	**Casirati S.**, Safeeq M., and Conklin M. H., (2020) "Influence of snow accumulation and ephemerality on recent tree mortality in the Sierra Nevada", Poster Presentation, Sierra Nevada Research Institute - UC Merced Research Week 2020: Research Contributions: "Addressing Climate Change for the Region and the World" - University of California Merced, California, USA
	Casirati S., Safeeq M., and Conklin M. H., (2020) "Influence of snow accumulation and ephemerality on recent tree mortality in the Sierra Nevada", Poster Presentation, Grad Visitation Poster Session: University of California Merced, California, USA
2019	**Casirati S.**, Safeeq M., and Conklin M. H., (2019) "Influence of snow accumulation and ephemerality on recent tree mortality in the Sierra Nevada", Poster Presentation, American Geophysical Union, Fall Meeting. San Francisco, California, USA
	Casirati S., Safeeq M., and Conklin M. H., (2019) "Influence of snow accumulation and ephemerality on recent tree mortality in the Sierra Nevada", Oral Presentation, Headwaters to Groundwater Annual Meeting, University of California Davis, California, USA
	Casirati S., Safeeq M., and Conklin M. H., (2019) "Influence of snow accumulation and ephemerality on recent tree mortality in the Sierra Nevada", Oral Presentation, Symposium: "Climate change and the ecology of Sierra Nevada forests", University of California Merced, California, USA

COMPETITIVE SCIENTIFIC PROPOSALS

(Pending)	NASA Early Career Investigator Program in Earth Science: Hydroclimatic Feedbacks and the Global Food-Water Nexus: Balancing Yield Targets with Sustainable Irrigation (Role: PI)

AWARDS and FELLOWSHIPS

2023	UC Merced ES Summer Fellowship ($7500)
2022	AGU FIHM Travel Grant ($1000)
2022	UC Merced ES Summer Fellowship ($7000)

2019 Fall-1819 UC Merced ES Bobcat Travel Fellowship ($1500)

TEACHING

University of California, Merced

2023 Field Methods in Snow Hydrology - ENVE 181 *(Teaching Assistant)*

2022 Surveying and Geomatics - CE010 *(Teaching Assistant)*

2022 Field methods in Snow Hydrology - ENVE 181 *(Teaching Assistant)*

2022 Calculus I - MATH 011 *(Teaching Assistant)*

2020 Spatial Analysis & Modeling – ENGR 180 *(Guest Lecture)*

MENTORING

2025- Chloe Faehndrich (UC Berkeley, Ph.D)

2024- Sarah Ryu (UC Berkeley, BS)

2020 Breanna Xiong (UC Davis, BS *Summer Intern California Ecosystem Climate Solutions project*)

ACADEMIC and PUBLIC SERVICE

2025 Panelist, NASA Grant Proposal Reviewer. Future Investigators in NASA Earth and Space Science and Technology (FINESST) Earth Panelist (Spring 2025)

2022 Equipment removal for the Hemlock Forest Restoration Project, Sierra Nevada Research Institute

2021 Environmental Systems seminar graduate student lead

2020 Environmental Systems seminar graduate student lead

2019 Weir repair activities at the Kings River Experimental Watershed, US Forest Service

Reviewer: *Journal of Hydrology, Journal of Advances in Modeling Earth Systems, Journal of Hydrometeorology, Theoretical and Applied Climatology, Review of Geophysics, Water Resources Research*

ADDITIONAL TRAINING

2021 Principle of Pedagogy course

2020 Processed-Based Hydrological modeling – University of Saskatchewan - Centre for Hydrology

2019 CUAHSI Master Class: Advanced Techniques in Watershed Science - University of Arizona - Biosphere 2

PROGRAMMING AND COMPUTING SKILLS

- **Operating systems:** Windows, Linux, Unix
- **Programming Languages:** R, Python, Fortran
- **Other computer tools:** ArcGIS, QGIS, CropWat, SWAT+, SUMMA, Google Earth Engine, NASA GEOS, NASA CLSM , LaTeX, Microsoft Office

LANGUAGES

- Fluent in Italian and English